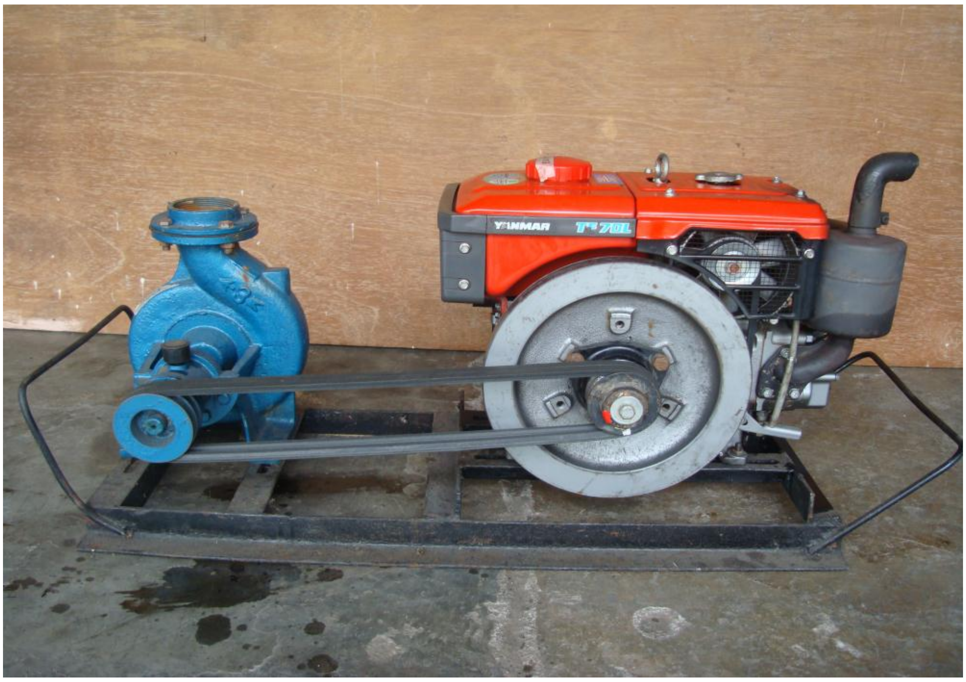
AMTEC

AGRICULTURAL MACHINERY TESTING AND EVALUATION CENTER

TEST REPORT NO. 2013 - 18



KITACO 3x3 MS coupled to YANMAR TF70L Diesel Engine

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KITACO 3 x 3 MS Centrifugal Pump coupled to a YANMAR TF70L Diesel Engine

Test Report No.	:	2013 - 18
Date of Test	:	March 6, 2013
Test Valid Until	:	March 6, 2018
Brand	:	KITACO
Model	:	3 x 3 MS
Serial No.	:	No data available
Type and Size	:	Single-end suction, 75mm x 75mm, Centrifugal pump
Manufacturer	:	Made in China
Test Requested by	:	EQUITY MACHINERIES, INC. National Highway, Cauayan City, Isabela
Machine Classification	:	Commercial

SUMMARY

The KITACO 3 x 3 MS centrifugal pump coupled to YANMAR TF70L diesel engine was tested for performance and suction condition. It was tested at the recommended full throttle speed setting of the engine. The engine and pump used double groove B-section pulleys with actual diameters of 89 mm (3.50 inches) and 101.6 mm (4.0 inches) and two pieces of V-belts. The engine and pump were mounted on a steel base.

At zero discharge condition, the Maximum Total Head (H) was 21.1 m. The shaft speeds of the pump and engine at this point were 2267 and 2513 rpm, respectively, with Engine Fuel Consumption Rate of 0.88 L/h. The Maximum Discharge Capacity of 14.8 L/s occurred at 9.11 m Total Head and shaft speeds of 2155 rpm and 2466 rpm for the pump and engine, respectively. The Fuel Consumption Rate at this point was 1.57 L/h.

The Maximum Fuel Consumption Rate of 1.57 L/h occurred at the point of Maximum Discharge Capacity.

The suction condition of the pump was also tested. With zero discharge, the Maximum Total Suction Lift was 8.19 m. The Net Positive Suction Head Available at this point was 1.70 m.

PURPOSE AND SCOPE OF TEST

The purpose of the test was to establish the pumpset performance which included the Fuel Consumption Rate and the Total Head at varying discharge capacities and suction conditions. **Figure 1** shows the pump mounted on the laboratory test set-up.

The test was conducted on March 6, 2013 at the AMTEC Test Laboratory, UPLB, College, Laguna.

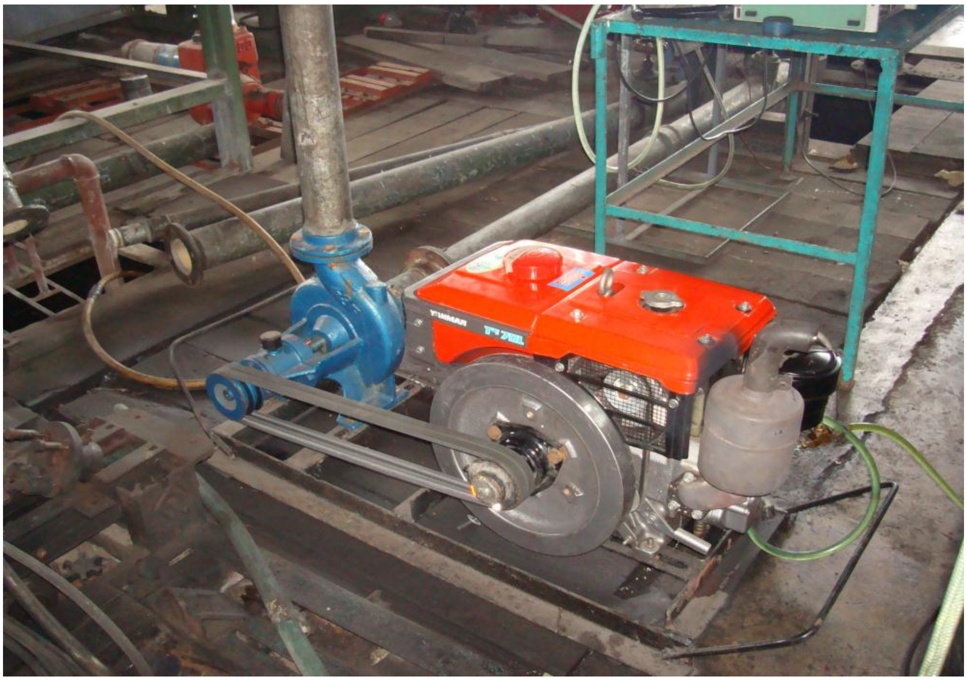

Figure 1. KITACO 3 x 3 MS Centrifugal pump coupled to YANMAR TF70L Diesel engine on the test set-up

METHOD OF TEST

The pump was tested following the " Philippine Agricultural Engineering Standards, PAES 115:2000, Agricultural Machinery – Centrifugal, Mixed-Flow, Axial-Flow Water Pumps – Methods of Test".

DESCRIPTION OF THE PUMP

The KITACO 3 x 3 MS Centrifugal pump coupled to YANMAR TF70L Diesel Engine shown in **Figure 2**, was a single-end suction, 75 mm x 75 mm, centrifugal pump coupled to a 5.22 kW YANMAR TF70L water-cooled diesel engine. The pump and engine were mounted on a steel base. The power from the engine was transmitted to the pump shaft through a double groove B-section pulley and V-belts. The actual diameter of the engine and pump pulleys were 89 mm (3.50 inches) and 101.6 mm, respectively. The suction inlet and discharge outlet were threaded flanges and designed for a 75 mm diameter pipe. The impeller was a closed-type made of brass material connected to the pump shaft through a thread. It was housed in a cast iron volute casing. The priming inlet was located on top of the pump beside the discharge outlet. The discharge outlet was located on top of the pump facing upward. The stuffing box used a mechanical seal. The pump shaft rotated on ball bearings kept in place by a bearing housing bolted to the casing.

The primemover, a YANMAR TF70L Diesel engine, was a 5.22 kW (7.0 hp), four-stroke, water-cooled, single horizontal cylinder diesel engine. Fuel feed system was gravity type with the fuel tank located on top of the engine. It utilized oil bath type air cleaner. Lubrication system was force-feed type. Cooling was done through the circulation of water from the cylinder jacket to the radiator. It was started by a hand crank system. A muffler was used in the exhaust system.

Table 1 shows the detailed specifications of the pump.

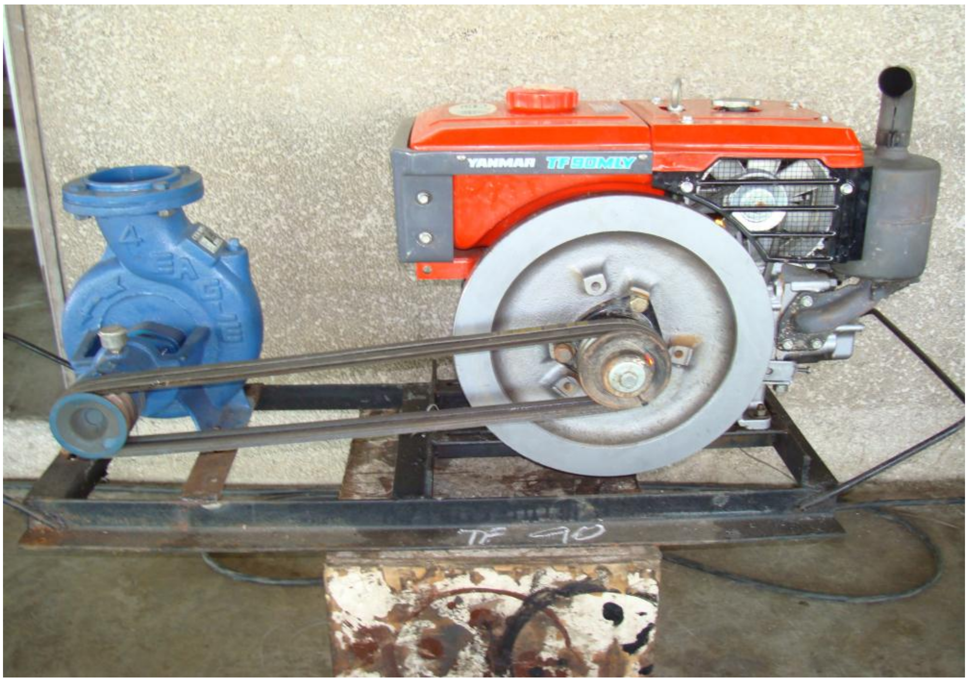

Figure 2. KITACO 3 x 3 MS Centrifugal Pump coupled to a YANMAR TF70L Diesel Engine

PUMPSET SPECIFICATIONS

Table 1. Detailed specifications of the KITACO 3 x 3 MS Centrifugal Pump coupled to a YANMAR TF70L Diesel Engine

1.	Brand		:	KITACO
2.	Model		:	3 x 3 MS
3.	Serial No.		:	No data available
4.	Make		:	Made in China
5.	Type		:	Single-end suction, Centrifugal pump
6.	Size (mm)		:	75 x 75
7.	Overall Dimension and Weight (Including mounting frame and engine)			
	7.1	Length (mm)	:	1220
	7.2	Width (mm)	:	460
	7.3	Height (mm)	:	520
	7.4	Weight (kg)	:	140.0
8.	Impeller			
	8.1	Type	:	Closed
	8.2	Diameter (mm)	:	174
	8.3	Width (mm)	:	20
	8.4	No. of vanes	:	8
	8.5	Material	:	Brass
9.	Suction Inlet			
	9.1	Type	:	Threaded flange
	9.2	Diameter (mm)	:	100
	9.3	Material	:	Cast Iron

Table 1. (Cont'n)

10.	Discharge Outlet		
	10.1 Type	:	Threaded flange
	10.2 Diameter (mm)	:	100
	10.3 Material	:	Cast Iron
11.	Casing		
	11.1 Type	:	Volute
	11.2 Material	:	Cast Iron
12	Stuffing box	:	Mechanical seal
13.	Rotation	:	Clockwise when facing the suction inlet
14.	Rated speed (rpm)	:	No data available
15.	Rated Maximum Discharge Capacity (L/s)	:	No data available
16.	Rated Maximum Total Head (m)	:	No data available
17.	Primemover		
	17.1 Brand	:	YANMAR
	17.2 Model	:	TF70L
	17.3 Serial No.	:	DC2114
	17.4 Make	:	YANMAR CO., LTD. Indonesia
	17.5 Type	:	Four stroke, water-cooled single horizontal cylinder diesel engine
	17.6 Rated Output Power (kW)		
	Maximum	:	5.22 @ 2400 rpm
	Continuous	:	4.48 @ 2400 rpm

Table 1. (Cont'n)

17.8	Bore x Stroke	:	78 x 80
17.9	Displacement Volume (cc)	:	382
17.10	Cooling System	:	Water-cooled
17.11	Lubricating System	:	Force-feed type
17.12	Starting System	:	Manual cranking lever
17.13	Air Cleaner	:	Oil bath
17.14	Exhaust System	:	Muffler

PERFORMANCE TEST

Results of performance test are shown in **Table 2** and plotted in **Figure 3**.

Pump Tested : KITACO 3 x 3 MS coupled to YANMAR TF70L Diesel engine

Date of Test : March 6, 2013

Test Condition

Ambient Temperature (^{o}C) : 30.0
Atmospheric Pressure (mb) : 1013
Pumping Water Temperature (^{o}C) : 29.8

Table 2. Results of Performance Test

ENGINE SHAFT SPEED (rpm)	PUMP SHAFT SPEED (rpm)	FUEL CONSUMP-TION (L/h)	TOTAL HEAD (m)	DISCHARGE CAPACITY (L/s)	WATER POWER (kW)	SYSTEM EFFICIENCY (%)
2513	2267	0.88	21.1	0	0	0
2496	2223	1.28	20.9	6.67	1.37	10.2
2489	2206	1.35	19.5	7.76	1.49	10.6
2482	2192	1.42	18.6	9.44	1.72	11.6
2474	2180	1.46	17.5	10.6	1.81	11.9
2471	2174	1.48	16.5	11.6	1.87	12.2
2464	2163	1.53	15.5	12.6	1.91	12.0
2459	2150	1.55	13.8	14.0	1.88	11.7
2462	2152	1.56	12.1	14.7	1.74	10.6
2466	2155	1.57	9.11	14.8	1.32	8.07

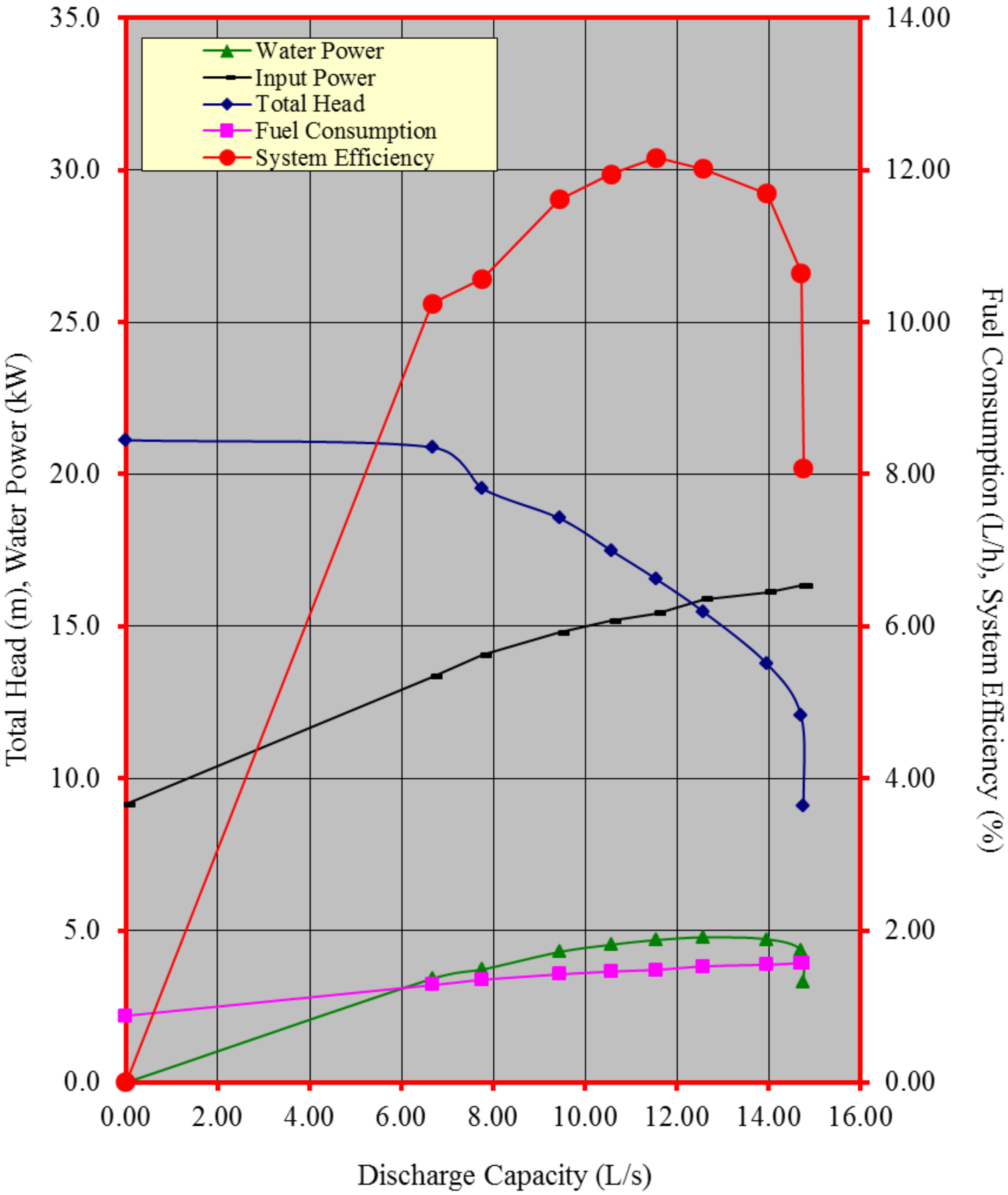

Figure 3. Performance characteristic curves of the KITACO 3 x 3 MS centrifugal pump coupled to YANMAR TF70L Diesel Engine

CAVITATION TEST

Results of Cavitation Test are shown in **Table 3**.

Pump Tested	:	KITACO 3 x 3 MS Pump coupled to YANMAR TF70L Diesel engine
Date of Test	:	March 6, 2013
Test Condition		
Ambient Temperature (^{o}C)	:	29.6
Atmospheric Pressure (mb)	:	1013
Pumping Water Temperature (^{o}C)	:	30.0

Table 3. Results of Cavitation Test

ENGINE SHAFT SPEED (rpm)	PUMP SHAFT SPEED (rpm)	TOTAL SUCTION LIFT (m)	NET POSITIVE SUCTION HEAD AVAILABLE (m)	TOTAL HEAD (m)	DISCHARGE CAPACITY (L/s)
2470	2148	2.62	7.27	9.03	22.2
2478	2161	3.65	6.24	9.62	20.2
2500	2202	5.53	4.36	10.5	14.0
2508	2218	6.41	3.48	10.7	10.8
2512	2242	7.32	2.57	11.3	4.78
2520	2253	8.19	1.70	12.0	0

DATA COMPUTATION

ANNEX 1. Formula used during calculation and testing

1. Total Head (m), H

$$TDH = H_d - H_s$$

Where: $$H_d = \frac{P_2}{\gamma} + \frac{V_2^2}{2g} + Z_2$$

$$H_s = \frac{P_1}{\gamma} + \frac{V_1^2}{2g} + Z_1$$

H_d = Total Discharge Head (m)

H_s = Total Suction Lift/Head (m)

$\frac{P_2}{\gamma}$ = Pressure gage reading on the discharge pipe at gage connection converted to meter

$\frac{P_1}{\gamma}$ = Vacuum gage reading on the suction pipe at the point of gage connection converted to meter

$\frac{V_2^2}{2g}$ = Velocity head on the discharge pipe converted to meter. Computed by determining the velocity of liquid on the pipe by formula, V = Q/A where V, velocity of liquid inside the pipe in meter per per second, Q, Discharge rate of the pump in liters per second, and A, Cross-sectional area of the pipe in square meter.

$\frac{V_1^2}{2g}$ = Velocity head on the suction pipe converted to meter

Annex (Cont'n)

Z_2 = The distance from the center of the pressure gage face to the centerline of the pump in meters

Z_1 = The distance from the center of the vacuum gage face to the centerline of the pump in meters

2. Discharge Capacity,Q, in liters per second (L/s) is determined through gravimetric method using a platform scale and a stopwatch.

3. Water Power,WP, is the output power of the pump in kilowatt (kW) and computed using:

$$WP = \frac{H \, X \, Q}{102}$$

4. Input Power , IP, is the power on the input shaft of the pump. This is the power in kilowatt required to drive the pump and is determine by the formula:

$$IP = \frac{T \, X \, N}{974}$$

Where:

T = Torque in kilogram-meter, on the input shaft of the pump and determined by a dynamometer

N = Input shaft angular speed in RPM

5. Pump Efficiency, η,

$$\eta = \frac{WP}{IP}$$

6. Net Positive Suction Head Available, NPSHA. This is the net force (converted to meter) needed to push the water inside the pipe and computed by the formula:

$$NPSHA = P_{atm} - P_{vp} - H_s$$

Where:

P_{atm} = Atmospheric pressure during the test converted to meters

P_{vp} = Vapor pressure of the test liquid at certain temperature converted to meters

Test Engineer:

ROMULO E. EUSEBIO

Checked by:

DARWIN C. ARANGUREN

Approved for release:

DELFIN C. SUMINISTRADO
Director

Date

N.B.

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